

BREEDIFY

AI-Powered Cattle & Buffalo Breed Classification

Tech Stack Document

Complete Technical Architecture & Stack Reference

Version 1.0.0

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1. High-Level Architecture

BREEDIFY follows a modern decoupled architecture with a React Single Page Application (SPA) frontend communicating via REST APIs with a Python FastAPI backend. The backend orchestrates AI/ML services including Google Gemini for breed classification and a custom iris biometric pipeline for animal identity verification.

Architecture Overview

Component	Technology
Client (Browser)	React 18 SPA + Vite + Tailwind CSS + React Router v6 + i18next
API Gateway	Vite dev proxy (/api/* and /uploads/*) forwarding to FastAPI at port 8000
Backend Server	FastAPI + Uvicorn serving REST API endpoints
AI Services	Vision Pipeline (Gemini multimodal) + AI Agent (Gemini LLM + fallback KB)
Biometrics	Custom iris recognition pipeline: detect > segment > normalize > encode > match
Database	SQLite (bovine.db) with 10 tables
External	Google Gemini API (vision + chat), Google Fonts CDN

2. Frontend Stack

Technology	Version	Layer	Purpose
React	18.3.1	UI Framework	Component-based Single Page Application
Vite	5.4.2	Build Tool	Dev server, HMR, production bundling
React Router DOM	6.26.2	Routing	Client-side routing with nested layouts
Tailwind CSS	3.4.10	Styling	Utility-first CSS framework
PostCSS + Autoprefixer	8.4.45 / 10.4.20	CSS Processing	Vendor prefixes, Tailwind compilation
i18next + react-i18next	23.15.1 / 15.0.1	i18n	Multi-language support (9 languages)
Lucide React	0.439.0	Icons	SVG icon library
clsx + tailwind-merge	2.1.1 / 2.5.2	CSS Utilities	Conditional class composition
jsPDF	4.2.1	PDF Export	Client-side PDF generation
Google Fonts	-	Typography	Playfair Display (serif) + Inter (sans)

Frontend Directory Structure

Path	Description
src/App.jsx	Root component + all route definitions (387 lines)
src/main.jsx	React DOM entry point
src/index.css	Global styles + Tailwind directives
src/i18n.js	i18next initialization with 9 languages
src/components/ (4 files)	AppLayout, ErrorState, EyeCaptureModal, MarketplaceSubNav
src/pages/ (27 files)	All page components (Login, Dashboard, Scanner, Marketplace, etc.)
src/locales/ (9 files)	Language JSON files (en, hi, te, gu, ta, kn, mr, bn, pa)
src/utills/ (5 files)	RBAC, marketplace/passport storage, PDF export
src/data/ (1 file)	Static breed reference data (sampleBreeds.js)

Page Components (27 total)

Component	Description
Login.jsx	Auth / role selection (Farmer, Vet, Vendor)
FarmerDashboard.jsx	Farmer portal home with marketplace, passports, quick actions
VetDashboard.jsx	Veterinarian portal home with review queue
VendorDashboard.jsx	Milk vendor portal with collection routes
Scanner.jsx	AI breed scan camera interface
ScanResult.jsx	Full scan results with AI Agent, passport creation, feedback

Marketplace.jsx	Livestock marketplace browse with filters
SellAnimal.jsx	Multi-step form to list animal for sale
AnimalDetail.jsx	Single animal marketplace detail page
MyListings.jsx	User's active marketplace listings
SavedAnimals.jsx	Favorited/saved animals
MyPurchases.jsx	Purchase history
SalesHistory.jsx	Sales transaction history
DigitalPassport.jsx	Animal passport registry (vet-gated creation)
PassportDetail.jsx	Individual passport view with health records
Identity.jsx	Iris biometric identity system
Health.jsx	Health records & diagnostics
Assistant.jsx	BREEDIFY AI Agent chat interface
InsuranceFinance.jsx	Insurance providers & government schemes
History.jsx	Scan history log
Reports.jsx	Analytics and reports
Profile.jsx	User profile management
Settings.jsx	App settings
CommunityFeedback.jsx	HITL breed verification feedback
AppPreview.jsx	App preview / landing page
ErrorPage.jsx	404 / error page
Dashboard.jsx	Role-based dashboard router

3. Backend Stack

Technology	Version	Layer	Purpose
FastAPI	latest	Web Framework	Async REST API framework with auto docs
Uvicorn	latest	ASGI Server	High-performance ASGI server
SQLite	built-in	Database	Embedded relational database
Pillow (PIL)	latest	Image Processing	Image analysis, resizing, quality checks
Google Gemini API	v1beta	AI / LLM	Breed classification + AI Agent chat
Requests	latest	HTTP Client	External API communication
Python	3.x	Language	Server-side runtime

Backend Directory Structure

File	Size	Description
server/main.py	696 lines	FastAPI app + all 18 API routes
server/vision_pipeline.py	626 lines	AI breed classification pipeline (Gemini vision)
server/copilot_service.py	212 lines	BREEDIFY AI Agent / Gemini chat service
server/database.py	413 lines	SQLite schema, init, migrations, seed data
server/bovine.db	200 KB	SQLite database file
server/breeds_dump.json	12 KB	Breed knowledge base export
server/test_iris_pipeline.py	9 KB	Iris biometric pipeline test suite
server/uploads/	-	Uploaded scan images directory

Iris Biometric Subsystem (server/iris/)

Module	Size	Purpose
iris_service.py	23.9 KB	Core biometric orchestration service
iris_detector.py	5.2 KB	Eye/iris region detection
iris_segmenter.py	3.6 KB	Iris boundary segmentation (pupil + limbus)
iris_normalizer.py	3.1 KB	Daugman rubber-sheet normalization
iris_encoder.py	3.9 KB	IrisCode generation (Gabor filter bank)
iris_matcher.py	5.7 KB	Hamming distance matching with rotation
iris_quality.py	3.9 KB	Blur, occlusion, exposure assessment
iris_config.py	2.3 KB	Configuration constants
bpa_repository.py	3.3 KB	BPA (Pashu Aadhaar) demo data

4. AI / ML Pipeline

4.1 Breed Classification (Vision Pipeline)

The vision pipeline accepts uploaded cattle/buffalo images and processes them through Google Gemini multimodal AI for breed identification, age estimation, pregnancy detection, and morphological feature extraction.

Stage	Description
1. Image Upload	User uploads full-body photo (max 10 MB)
2. Quality Check	Pillow validates resolution, blur, exposure
3. Gemini Vision API	Multimodal analysis for breed, species, features
4. Breed Classification	Identifies breed from 19 supported breeds + confidence score
5. Age Estimation	Horn ring morphology, dentition analysis, body frame maturity
6. Pregnancy Detection	Visual body conformation analysis
7. Nearest Matches	Pre-computed breed similarity graph (phenotypic distance)
8. Structured Result	Full result JSON with geotag, features, breed data

Supported Breeds (19 total)

Cattle Breeds (13):

Gir, Sahiwal, Red Sindhi, Tharparkar, Kankrej, Ongole, Hallikar, Hariana, Kangayam, Rath, Deoni, Jersey Cross, HF Cross

Buffalo Breeds (6):

Murrah, Nili Ravi, Jaffarabadi, Surti, Mehsana, Banni

4.2 Iris Biometric Identity System

A complete 6-stage iris biometric pipeline for unique animal identification, compliant with BPA (Bovine Pashu Aadhaar) standards. Supports calf-to-adult lifecycle re-enrollment.

Stage	Module	Technique
1. Detection	iris_detector.py	Eye region localization from uploaded image
2. Segmentation	iris_segmenter.py	Pupil and limbus boundary detection
3. Normalization	iris_normalizer.py	Daugman rubber-sheet unwrapping to rectangular
4. Encoding	iris_encoder.py	Gabor filter bank produces binary IrisCode
5. Matching	iris_matcher.py	Hamming distance with rotation compensation
6. Quality	iris_quality.py	Blur, occlusion, exposure quality scoring

4.3 AI Agent (Chat / Q&A)

Feature	Detail
Primary Model	Google Gemini (gemini-flash-latest, gemini-3.1-flash-lite)
Fallback	Built-in expert knowledge engine (rule-based clinical KB)
Persona	ICAR-NDRI veterinary specialist
Knowledge	Bloat, Mastitis, Fever, Diarrhea, LSD, Vaccination, PKCC Loans, Insurance, AI timing
Integration	Sidebar chat page + embedded on scan result page with breed-specific queries
Temperature	0.3 (factual, low hallucination)
Max Tokens	800 per response

5. Database Schema (SQLite)

The application uses SQLite (bovine.db) with 10 tables covering users, breeds, animals, scans, feedback, insurance, schemes, market prices, and iris biometric records.

Table	Key Columns	Description
users	id, name, mobile, location, role, preferred_language, created_at	Registered users (farmers, vets, vendors)
breeds	id, name, species, origin, avg_milk_yield, lactation_period, cost_range, ideal_temp_range, feeding_notes, lifespan, description, key_features (JSON), disease_risks (JSON), image_url	Breed knowledge base (19 breeds)
animals	id, owner_id, tag_number, sex, predicted_breed_id, confirmed_breed_id, last_scan_id	Registered animals
scans	id, animal_id, user_id, images, predicted_breed, confidence, pregnancy_status, prediction_status, geotag, age_range,	AI scan records
feedback	id, scan_id, submitted_by, verdict, suggested_breed, note, reviewed_by, review_status	HITL breed verification feedback
insurance_providers	id, name, covered_species, region, coverage_amount, premium_rate, contact	Livestock insurance providers
schemes	id, type, name, offered_by, eligibility, amount_range, link	Government schemes & loans
mandi_prices	id, state, district, item, price_per_unit, date	Market (mandi) prices
animal_identities	id, bpa_tag_id, animal_type, breed, age_months, biometric_status	BPA / Pashu Aadhaar identity records
biometric_templates	id, animal_identity_id, template_version, eye_side, iris_template, quality_score	Encoded iris biometric templates

6. REST API Endpoints (18 total)

Authentication

Method	Endpoint	Description
POST	/api/auth/login	User login (name, mobile, role, language)

AI Breed Scanning

Method	Endpoint	Description
POST	/api/scan	Upload image -> breed classification pipeline
GET	/api/scans	List all scan history
GET	/api/scans/:scanId	Get single scan result with full details
POST	/api/scan/:scanId/feedback	Submit HITL breed verification feedback
POST	/api/scan/:scanId/transfer-ownership	Transfer animal ownership
POST	/api/scan/:scanId/geotag	Update scan geolocation

Breeds & AI Agent

Method	Endpoint	Description
GET	/api/breeds	List all 19 breeds with full details
GET	/api/breeds/:breedId	Get single breed information
POST	/api/chat	Chat with BREEDIFY AI Agent (Gemini + fallback KB)

Veterinary Review

Method	Endpoint	Description
GET	/api/vet/queue	Get low-confidence scans for vet review
POST	/api/vet/review	Vet confirm/correct breed prediction

Iris Biometric Identity

Method	Endpoint	Description
POST	/api/identity/register	Register animal + enroll initial iris template
POST	/api/identity/verify	Verify animal identity against enrolled templates
POST	/api/identity/update-biometric	Calf-to-Adult lifecycle re-enrollment
POST	/api/identity/check-duplicate	Screen for duplicate iris records
GET	/api/identity/:bpaTagId	Get animal identity + biometric history

GET	/api/identity/demo/animals	List demo BPA records
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Insurance, Schemes & Market

Method	Endpoint	Description
GET	/api/insurance	List livestock insurance providers
GET	/api/schemes	List government schemes & loans
GET	/api/mandi	Get mandi (market) prices
POST	/api/flw/batch-sync	Batch sync offline scan records
GET	/api/health/status	Health check

7. Design System

Color Palette

Token	Hex	Usage
cream-50	#FAF8F3	Page background
cream-100	#F7F3EA	Card backgrounds
cream-200	#EFE8D8	Borders, dividers
cream-300	#E3D7BD	Muted accents
forest-900	#1A241B	Sidebar background
forest-800	#222E23	Sidebar hover
forest-700	#2F3D30	Dark text on light
forest-600	#3A4A3C	Secondary dark
forest-500	#4B5E4E	Muted dark
forest-100	#E5EBE6	Light green tint
terracotta	#C1552E	Primary CTA / accent
terracotta-hover	#A84723	Button hover state
terracotta-light	#F8EDE8	Light accent background
terracotta-dark	#8C381A	Active / pressed
charcoal	#2A2A28	Body text
charcoal-muted	#8A8A80	Secondary text
charcoal-light	#5A5A54	Tertiary text

Typography

Role	Font Family	Weights
Headings	Playfair Display (serif)	500, 600, 700, 800
Body Text	Inter (sans-serif)	300, 400, 500, 600, 700

Shadow System

Token	CSS Value	Usage
soft	0 4px 20px -2px rgba(42,42,40, 0.05)	Subtle elevation
card	0 10px 30px -5px rgba(42,42,40, 0.08)	Card elevation
glow	0 0 25px rgba(193,85,46, 0.25)	CTA glow effect

8. Role-Based Access Control (RBAC)

The application supports three user roles, each with distinct permissions enforced both on the frontend (UI gating) and referenced in the backend.

Farmer (role: "farmer")

- animal:create

Veterinarian (role: "vet")

- animal:read:assigned

Milk Vendor (role: "vendor")

- milk:create

IMPORTANT

Passport creation (passport:create) is restricted exclusively to registered veterinarians. Farmers can view their own passports but cannot create or register new animal passport records. This complies with INAPH veterinary regulations.

9. Internationalization (i18n)

The application supports 9 Indian languages using i18next with React integration. Language preference is persisted to localStorage and applied globally across all UI components.

Code	Language	Script	File Size	Coverage
en	English	Latin	21 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
hi	Hindi	Devanagari	37 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
te	Telugu	Telugu	39 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
gu	Gujarati	Gujarati	34 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
ta	Tamil	Tamil	27 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
kn	Kannada	Kannada	26 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
mr	Marathi	Devanagari	25 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
bn	Bengali	Bengali	26 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings
pa	Punjabi	Gurmukhi	25 KB	Navigation, dashboard, marketplace, scanner, passports, AI agent, health, settings

10. Client-Side Data Layer

The marketplace and passport features use localStorage for client-side persistence in this prototype version. The data layer is abstracted through utility modules that can be swapped for API-backed implementations.

Module	Storage	Purpose
marketplaceStorage.js	localStorage	Marketplace listings, saved animals, purchases, sales history
passportStorage.js	localStorage	Digital animal passports, health records
ecosystemStorage.js	localStorage	Dashboard ecosystem data and stats
pdfExport.js	jsPDF	Client-side PDF generation for passports and reports

11. Development & Build

Development Commands

Command	Description
npm run dev	Start Vite dev server on port 5173 with HMR
python server/main.py	Start FastAPI server on port 8000 via Uvicorn
npm run build	Vite production build to dist/
npm run preview	Preview production build locally

Vite Proxy Configuration

The Vite dev server proxies /api/* and /uploads/* requests to the FastAPI backend at http://127.0.0.1:8000, enabling seamless frontend-backend communication during development without CORS issues.

Production Build Output

File	Raw Size	Gzipped	Type
index.html	1.16 KB	0.66 KB	HTML
index-*.css	63.06 KB	10.63 KB	CSS
index-*.js	1,362 KB	391 KB	Main JS bundle
html2canvas.esm-*.js	201 KB	48 KB	Canvas lib
purify.es-*.js	28.9 KB	11.1 KB	DOMPurify

12. Static Assets

Path	Contents
public/breedify_logo.jpg	Application logo and favicon (163 KB)
public/logo.png	Alternate logo format (163 KB)
public/breeds/ (20 images)	Realistic breed photographs for all 19 breeds + calf
public/preview/	App preview / marketing screenshots
public/stickers/	Decorative stickers and badges

13. External Dependencies

Service	Type	Purpose
Google Gemini API	Cloud AI	Breed classification (multimodal vision), age estimation, AI Agent chat
Google Fonts	CDN	Playfair Display + Inter font loading
Unsplash	CDN	Fallback breed reference images in database seed data

14. Summary

Dimension	Detail
Frontend	React 18 + Vite 5 + Tailwind CSS 3 + React Router v6
Backend	Python FastAPI + Uvicorn + SQLite
AI Engine	Google Gemini (multimodal vision + LLM chat)
Biometrics	Custom iris recognition pipeline (6-stage)
Languages	9 Indian languages (en, hi, te, gu, ta, kn, mr, bn, pa)
User Roles	Farmer, Veterinarian, Milk Vendor
Supported Breeds	19 (13 cattle + 6 buffalo)
API Endpoints	18 REST endpoints
Database Tables	10 SQLite tables
Page Components	27 React page components
Build Tool	Vite 5.4.2 with React plugin
CSS Framework	Tailwind CSS 3.4.10 with custom design tokens
State Management	React useState/useEffect + localStorage
PDF Generation	jsPDF (client-side)
Icon Library	Lucide React (SVG)

BREEDIFY v1.0.0

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